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NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Seventh Semester B.E. (Credit System) Degree Examinations
Supplementary Examinations - December 2025

21EC8X18 - CONSUMER ELECTRONICS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) Describe the construction and working of moving coil (dynamic) microphone with relevant diagram.	8	L*2	1	1
	b) Explain the working of crystal microphone with relevant diagram.	8	L2	1	1
	c) Define doppler effect and beats relating to acoustic signal.	4	L1	1	1
2.	a) Explain the construction and working of ribbon microphone with necessary diagram.	8	L2	2	1
	b) Define the term aperture. Describe the components of camera.	8	L2	2	1
	c) Compare monophony and stereophony.	4	L2	1	1
3.	a) Summarize the relative merits and demerits of horn type and cone type speakers.	8	L2	1	1
	b) Explain the features for selecting CCTV camera.	4	L2	2	1
	c) Explain with diagram the working of a permanent magnet dynamic loudspeaker.	8	L2	1	1
Unit – II					
4.	a) Explain the working of different types of cartridges.	8	L2	3	1
	b) Explain the standard recording procedure with diagram.	8	L2	3	1
	c) Summarize the merits of DTH service.	4	L2	4	1
5.	a) Explain the working principle of superheterodyne receiver with relevant diagram.	8	L2	4	1
	b) Explain the working of variable area optical recording system with relevant diagram.	8	L2	3	1
	c) Explain the principles of uniselector used in snitching systems.	4	L2	4	1
6.	a) Explain digital recording using laser technology with relevant block diagram.	8	L2	3	1
	b) Explain the various components of earth station and satellite station.	6	L2	4	1
	c) Explain the three segments of GPS elements.	6	L2	4	1
Unit – III					
7.	a) Explain four processes of refrigeration system with relevant block diagram.	8	L2	5	1
	b) Explain the working principle of microwave oven with relevant functional diagram.	8	L2	5	1
	c) Explain the features of ATM.	4	L2	5	1
8.	a) Explain the working principle of washing machine with relevant program flowchart.	8	L2	5	1
	b) What is nanotechnology? Write a short note on carbon nanotubes.	6	L2	5	1
	c) Summarize the importance of electronics in healthcare systems.	6	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
